

Final Project: Details

For your final project for Data in Action, you will work in small groups to explore and analyze a dataset using the data science skills you have learned this week.

Groups will be assigned based on dataset preferences; no groups will be working on the same dataset. Once groups have been assigned, you will be provided with a CSV file of the dataset as well as an accompanying information sheet. This information sheet will include details such as where the data is from, why it was collected, and what columns there are. Each member of your group will be provided with a hard copy of this sheet as well as a pdf file of it. **Note: do NOT copy-paste any text from these info sheets for your final project presentations.**

Analysis Requirements

As a group, you will pose a research question, state a hypothesis, conduct an exploratory data analysis (EDA), and draw a conclusion. Below is a list of the parts that you are expected to complete.

- Research question
- Hypothesis
- x1 plot with one variable shown. Can be a dot plot or a histogram, your choice. Must show either the mean or median on the plot, you decide which is more appropriate.
- x1 plot with two variables shown (one on each axis). Can be a scatterplot, line graph, bar graph, etc.
- x1 plot with three or more variables shown (one on each axis + variable(s) mapped to color, size, etc.)
- All plots must have nice axis labels and titles

All analyses must be completed using CODAP. Your dataset will be provided to you as a CSV file that you can then import into CODAP. Although will all work individually in your own CODAP file, please be sure to communicate and collaborate throughout the process as you will ultimately need to agree on which plots to present.

Presentation Requirements

On the final day of camp (Friday, July 17th), you will give your final presentations. Each presentation will be **10 minutes** followed by 5 minutes for questions. You can split up the talking portions however you like, as long as everyone speaks an equal amount. Everyone in your group must speak.

Slides Outline

Your presentation must be made in Google slides. Please be sure to share your the document with everyone in your group, as well as with myself (gracelawley@lclark.edu) before the presentations. Below is an outline for what to include in your slides. Keep in mind that some sections will require more than one slide to cover fully.

1. Title slide

- Contains your project title, date, and the names of everyone in your group.

2. Background

- What is your topic? Remember, only your group will have seen your dataset and info sheet, so we are relying on you to tell us about it.
 - If you include information beyond what is provided on the background sheet given to you at the start of the project, be sure to cite your sources (url link at the bottom of the slide is okay).
 - **Do NOT copy-paste any text from the info sheets.**
 - Is there any additional context you should share so we can better understand your topic?
- 3. Dataset**
 - Show a screenshot of the dataset. To make sure that it is still readable when projected, only take a screenshot of a portion of the dataset, not the entire thing.
 - Tell us about the structure of the dataset. How many columns and rows are there?
 - What does each row represent?
 - Where did this dataset come from?
 - 4. Data dictionary**
 - For each column, tell us the name, description, variable type, example values, and units (if applicable)
 - 5. Research question**
 - What is your research question?
 - 6. Hypothesis**
 - What is your hypothesis?
 - 7. Results**
 - Can format this however you would like, but I suggest using separate slides for of your plots.
 - When showing your plots, explain what each and talk us through what is shown.
 - 8. Conclusion**
 - Do your findings support or not support your hypothesis? Walk the audience through your reasoning
 - 9. Limitations**
 - What are the limitations of your analysis?
 - Some examples of things to consider:
 - Is there less data for a specific group and more for the others?
 - Does your dataset only span a limited time frame?
 - Is additional information missing that would be useful to have?
 - 10. Reflection**
 - Some ideas:
 - How did your analysis go? What was straightforward? What was difficult? Was anything surprising
 - If you had more time, what would you have done differently?
 - Is there anything else that you would have explored?
 - Did you pivot and switch research questions at some point? Why?